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Ganghui ZHANG

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Education

Visiting student	**Department of Mathematics, University of Regensburg** **Host** : Prof. Harald Garcke	Jun. 2024 – now
Ph.D. Candidate	**Yau Mathematical Sciences Center, Tsinghua University** **Major:** Computational Mathematics **Supervisor:** Prof. Chunmei Su	Sept. 2019 – now
B.Sc.	**Shandong University** **Major:** Pure and Applied Mathematics	Sept. 2015 – Jun. 2019

Research interest

- Numerical Analysis and Scientific Computing
- Geometric Partial Differential Equations

Publications

1. W. Jiang, C. Su and **G. Zhang**, Stable BDF time discretization of BGN-based parametric finite element methods for geometric flows, SIAM J. Sci. Comput., 46 (2024), pp. A2874-A2898.
2. W. Jiang, C. Su and **G. Zhang**, A second-order in time, BGN-based parametric finite element method for geometric flows of curves, J. Comp. Phys., 514 (2024), article 113220.
3. W. Jiang, C. Su and **G. Zhang**, A convexity-preserving and perimeter-decreasing parametric finite element method for the area-preserving curve shortening flow, SIAM J. Numer. Anal. 61 (2023), pp. 1989–2010.

Preprints

1. H. Garcke, W. Jiang, C. Su and **G. Zhang**, Structure-preserving parametric finite element method for curve diffusion based on Lagrange multiplier approaches, arXiv: 2408.13443.
2. W. Jiang, C. Su and **G. Zhang**, Convergence analysis of three semi-discrete numerical schemes for nonlocal geometric flows including perimeter terms, arXiv:2405.07690.

Minisymposium talks

1. Structure-preserving parametric finite element method for curve diffusion based on Lagrange multiplier approaches, RTG IntComSin Annual Meeting, Ellwangen, September 30, 2024.
2. A convexity-preserving and perimeter-decreasing parametric finite element method for the area-preserving curve shortening flow, Annual Conference of CSIAM, Kunming, October 15, 2023.

Honor

Yau Mathematical Sciences Center Scholarship	Tsinghua University 2023
National Scholarship	Shandong University 2018

Other Skills

Programming languages: MATLAB, The Distributed and Unified Numerics Environment (DUNE) based on C++.

Technologies: LaTeX.